llmXive Automated Review of Kwai Keye-VL-2.0 Technical Report

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The manuscript makes several strong factual claims regarding the novelty and performance of the Kwai Keye-VL-2.0 model. While the technical depth is high, a few claims regarding "pioneering" status and "lossless" processing require tighter alignment with the provided evidence and citations.

First, the Introduction (Section 1) asserts that the authors "pioneer the application of Multimodal DeepSeek Sparse Attention (DSA) in the visual domain," citing deepseek2025v32. A review of the provided bibliography shows that deepseek2025v32 is a technical report for a text-only LLM (DeepSeek-V3.2). The paper does not cite any prior work that applied DSA to visual tokens, nor does it explicitly state that the cited paper *only* covered text. If the application of DSA to vision is indeed a novel contribution of this work, the claim is valid but should be phrased to distinguish between the *method* (from DeepSeek) and the *application domain* (novel here). If DSA for vision was previously explored, the "pioneer" claim is factually incorrect. The current phrasing risks overstating the novelty of the *method* itself rather than its *application*.

Second, the Abstract and Introduction repeatedly claim "lossless 256K context processing" and "losslessly" handling video contexts. However, Section 1.2 ("Unified Visual Encoding") explicitly details an "Adaptive Video Pixel Budget" where videos longer than 256s are processed with scaling factors of 0.125, 0.25, or 0.5. This mechanism inherently reduces the number of visual tokens (and thus information) for long videos to fit the context window. Describing this as "lossless" is factually inaccurate; it is "context-efficient" or "lossy compression with adaptive scaling." The term "lossless" should be removed or strictly qualified to apply only to the context window management, not the visual content itself.

Third, in Section 4.1 (Video Understanding), the text below Table 1 states: "Keye-VL-2.0 achieves best results on LongVideoBench and Video-MME-v2 accuracy." While the "ACC" row in Table 1 supports this, the table also includes a "Non-Lin" (Non-Linear) row for Video-MME-v2, where Keye-VL-2.0 scores 18.5/24.2, significantly lower than the Qwen3-VL 235B baseline (26.3/28.1). The claim of "best results" is ambiguous without specifying that it applies only to the "ACC" metric, potentially misleading readers about the model's performance on the full benchmark suite.

Finally, the paper relies on several future-dated citations (e.g., openai2026gpt55, anthropic2026claude, deepmind2026gemini). While consistent with the paper's 2026 timeline, the validity of claims comparing against these models depends entirely on the existence and stability of these future reports. The authors should ensure these references are accessible or clearly marked as "forthcoming" if they are not yet public.

Action items.

- [**science**] The claim that Keye-VL-2.0 'pioneers the application of Multimodal DeepSeek Sparse Attention (DSA) in the visual domain' (Introduction) is unsupported. The cited source (deepseek2025v32) describes DSA for text-only LLMs. The paper provides no evidence that the cited work or any prior art applied DSA to visual tokens, making the 'pioneer' claim potentially overstated without a specific citation for the visual adaptation.
- [**writing**] The abstract and Introduction claim 'lossless 256K context processing' and 'losslessly' handling video contexts. However, Section 1.2 explicitly describes an 'Adaptive Video Pixel Budget' with scaling factors (0.125, 0.25, 0.5, 1.0) for videos exceeding 256s.

This implies information reduction (subsampling) for long videos, contradicting the 'lossless' claim. The term should be qualified (e.g., 'lossless up to X duration' or 'context-efficient').

- [**writing**] Table 1 (Video Understanding) lists 'Video-MME-v2 Non-Lin' scores where Keye-VL-2.0 (18.5/24.2) is significantly lower than Qwen3-VL 235B (26.3/28.1), yet the text below the table states Keye-VL-2.0 'achieves best results on... Video-MME-v2 accuracy'. While the 'ACC' row supports this, the inclusion of the 'Non-Lin' row in the same table without clarification creates ambiguity about the 'best results' claim across the full benchmark suite.
- [**writing**] The citation 'openai2026gpt55' (Introduction) refers to a 2026 report for 'GPT-5.5'. Given the current date context of the paper (2026), this is a future-dated citation. While acceptable for a preprint in that timeline, the claim that this model 'demonstrates substantial progress' relies on a source that may not be publicly verifiable or stable at the time of review, requiring the authors to ensure the URL and data are accessible.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

Figure 1 is a clear and well-structured schematic that effectively communicates the four-stage pre-training pipeline. The visual hierarchy, parameter status indicators, and data scale metrics are legible and align perfectly with the provided caption.

Figure 2

Figure 2 is a clear and well-structured schematic that effectively visualizes the four-stage pre-training pipeline described in the caption. All stages, parameter statuses (ViT, Projector, LLM), sequence lengths, data scales, and data types are clearly labeled and legible, with no missing controls or misleading scales.

Figure 3

Figure 3 effectively illustrates the scene-wise dense captioning concept with a clear visual breakdown of a video timeline into four distinct scenes. The layout is uncluttered, and the detailed text descriptions for each scene align perfectly with the provided caption and the visual content shown in the filmstrip.

Figure 4

Figure 4 effectively illustrates cost reductions but uses inconsistent y-axis scales between the two subplots, which may mislead readers about the relative magnitude of savings. Minor legend improvements would enhance clarity.

Figure 5

Figure 5 is a clear and well-organized table summarizing the model's performance across various benchmarks. However, the caption's statement about 'Higher is better' could be more precise by explicitly confirming that all listed metrics follow this rule, and some benchmark names could be formatted for better readability.

Action items.

- [**science**] Figure 4: The 'Decode Cost' y-axis scale is inconsistent with the 'Prefill Cost' scale (0.0–3.2 vs 0.0–0.06), yet the visual height of the 'Full Attention' line is nearly identical in both plots. This creates a misleading visual impression that the absolute cost reduction is similar in both phases, whereas the caption and axis labels imply a much larger absolute saving in decode. The scales should be comparable or explicitly noted as different to avoid misinterpretation.
- [**writing**] Figure 4: The legend at the top right ('DSA' and 'Full Attention') is not directly connected to the lines in the plots; while colors match, adding a small line sample next to each label in the legend would improve clarity.
- [**science**] Figure 5: The caption states 'Higher is better unless otherwise specified,' but the benchmark 'Video-MME-v2 ACC (64 Frames/512 Frames)' lists 'ACC' (Accuracy), which is a percentage metric. The values (e.g., 35.3 / 42.4) are likely percentages, yet the caption's phrasing suggests a general rule that might confuse readers if some benchmarks are actually loss-based or require lower scores. The caption should explicitly list any benchmarks where 'Lower is better' to avoid ambiguity, or confirm all
- [**writing**] Figure 5: The benchmark name 'Video-MME-v2 ACC (64 Frames/512 Frames)' is split across two lines in a way that breaks the logical grouping of the metric name and its configuration. It should be formatted to keep 'Video-MME-v2 ACC' together or clearly indicate the frame configurations as sub-rows or a single continuous label.
- [**writing**] Figure 5: The benchmark 'VideoMMMU' is listed without a clear definition of what 'MMMU' stands for in the context of video (e.g., is it a video adaptation of the MMMU benchmark?). While the caption mentions 'detailed benchmark descriptions... provided in the subsections below,' the figure itself should ideally include a footnote or abbreviation key for clarity.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript suffers from significant jargon overuse, frequently introducing complex acronyms and proprietary terms without definition, which alienates non-specialist readers. The Abstract immediately fails to define 'DSA' (DeepSeek Sparse Attention) and 'MOPD' (Cross-Modal Multi-Teacher On-Policy Distillation), presenting them as standard terms rather than specific architectural choices. This pattern continues throughout the text: 'GSPO' (Section 4.2.2), 'SPRR' (Section 4.2.3), and 'ExtraIO' (Section 5.1) are all used without expansion or brief explanation.

Furthermore, the paper relies on opaque internal naming conventions. Terms like 'Reception' and 'Remake' (Section 3.2) are used as verbs to describe data processing steps without clarifying what these actions entail. The model name 'Qwen3-30B-A3B-Thinking-2507' includes the suffix '2507', which appears to be a version date but is presented without context, confusing the reader about its significance. Even established external references like 'NaViT' are used without a brief reminder of the 'Patch n' Pack' mechanism they employ.

To make this technical report accessible to a broader audience, every acronym must be defined at its first occurrence, and proprietary or internal process names must be accompanied by a plain-language description of their function. The current density of undefined jargon creates an unnecessary barrier to understanding the paper's contributions.

Action items.

- [**science**] The manuscript suffers from significant jargon overuse, frequently introducing complex acronyms and proprietary terms without definition, which alienates non-specialist readers. The Abstract immediately fails to define 'DSA' (DeepSeek Sparse Attention) and 'MOPD' (Cross-Modal Multi-Teacher On-Policy Distillation), presenting them as standard terms rather than specific architectural choices. This pattern continues throughout the text: 'GSPO' (Section 4.2.2), 'SPRR' (Section 4.2.3), and 'ExtraIO'

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The manuscript presents a technically dense report on Kwai Keye-VL-2.0, but several logical inconsistencies and unsupported causal claims undermine the rigor of the conclusions.

First, the claim of "lossless 256K context processing" (Abstract) contradicts the described mechanism of DeepSeek Sparse Attention (DSA). Section 1.3 details a "Lightning Indexer" that selects a Top-k subset of tokens ($k = 2048$) for aggregation. By definition, attending to a sparse subset of the context implies information loss relative to full dense attention. The term "lossless" is logically inconsistent with the sparsification mechanism unless it refers solely to the *capacity* to process long sequences without truncation, not the fidelity of the attention mechanism. This ambiguity requires clarification to avoid misleading readers about the model's information retention capabilities.

Second, the complexity reduction claim in Section 1.3 ("reduces complexity from $O(L^2)$ to $O(Lk)$") is not fully supported by the provided equations. Equation 1 describes the computation of global index scores $I_{t,s}$, which involves interactions between query q_t and keys k_s. If this indexing step is computed for all s in the sequence, it inherently retains $O(L^2)$ complexity or requires a separate approximation not detailed in the text. The paper asserts the reduction but does not logically demonstrate how the indexing phase itself avoids the quadratic bottleneck, creating a gap between the premise (the mechanism) and the conclusion (the complexity reduction).

Third, the causal claim in Section 3.1 that Video RL "Improves benchmarks by 1%" is logically weak. The statement lacks specificity regarding which benchmarks were improved, the baseline model, and the statistical significance of the gain. A 1% improvement on a single metric does not logically support a general claim of benchmark improvement without further evidence.

Finally, the assertion that MOPD prevents "catastrophic forgetting" (Abstract, Section 1) is not logically derived from the described method in Section 3.2.4. The section details a distillation loss from multiple teachers but does not explicitly include mechanisms known to prevent forgetting (e.g., experience replay, specific regularization terms). Without these details, the conclusion that MOPD *specifically* prevents forgetting is an unsupported leap.

These issues require clarification to ensure the paper's conclusions follow logically from its premises.

Action items.

- **[writing]** The manuscript presents a technically dense report on Kwai Keye-VL-2.0, but several logical inconsistencies and unsupported causal claims undermine the rigor of the conclusions. First, the claim of "lossless 256K context processing" (Abstract) contradicts the described mechanism of DeepSeek Sparse Attention (DSA). Section 1.3 details a "Lightning Indexer" that selects a Top-k subset of tokens ($k = 2048$) for aggregation. By definition, attending to a sparse subset of the context implies informati

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The manuscript exhibits significant over-claiming regarding the novelty and capabilities of the proposed architecture, particularly in the Abstract and Introduction.

First, the term "lossless 256K context processing" (Abstract; Section 1) is technically inaccurate and misleading. The core mechanism described, DeepSeek Sparse Attention (DSA), relies on a "MQA-Style Lightning Indexer" to select a Top-k subset of tokens (Equation 1, Section 1.3). By definition, selecting a subset of tokens constitutes a lossy compression of the input sequence. While the model may retain *sufficient* information for high performance, claiming the process is "lossless" suggests that no information is discarded, which contradicts the fundamental operation of sparse attention. The authors should qualify this claim (e.g., "near-lossless" or "information-preserving") or explicitly define what "lossless" means in this context to avoid overstating the mechanism's fidelity.

Second, the Introduction claims the work "pioneers the application of Multimodal DeepSeek Sparse Attention (DSA) in the visual domain." This is a strong novelty claim that is not substantiated by the provided text. The paper cites DeepSeek-V3.2 (a text-only model) but offers no evidence that DSA has not been previously adapted for visual tokens in other research. Without a citation to prior art or a clear demonstration that this is the first such application, this statement constitutes an over-claim of novelty. The authors should either cite the first instance of this application or rephrase the claim to "we adapt DSA for multimodal contexts" to remain within the bounds of the evidence.

Finally, the description of Cross-Modal Multi-Teacher On-Policy Distillation (MOPD) as a solution that "robustly preserves" foundational reasoning and "effectively isolates" expertise (Introduction) overstates the empirical results. While the model performs well, the evaluation tables (Table 1, Table 2) show that Keye-VL-2.0 does not universally outperform all baselines (e.g., it trails Qwen3.5 on MLVU and GPT-5-mini on Video-MMMU). The text implies a complete resolution to the "Multimodal Alignment Dilemma," whereas the data suggests a strong but imperfect mitigation. The language should be tempered to reflect that the method *significantly reduces* catastrophic forgetting rather than eliminating it entirely.

Action items.

- **[writing]** The manuscript exhibits significant over-claiming regarding the novelty and capabilities of the proposed architecture, particularly in the Abstract and Introduction. First, the term "lossless 256K context processing" (Abstract; Section 1) is technically inaccurate and misleading. The core mechanism described, DeepSeek Sparse Attention (DSA), relies on a "MQA-Style Lightning Indexer" to select a Top-k subset of tokens (Equation 1, Section 1.3). By definition, selecting a subset of tokens constitu

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The manuscript presents a technically advanced multimodal model with significant capabilities in long-video understanding and autonomous agent execution. From a safety and ethics perspective, the paper currently lacks sufficient disclosure regarding data provenance, privacy safeguards, and potential dual-use risks associated with its agentic features.

First, regarding **data privacy and consent**, the paper states the use of massive web-scraped datasets (LAION, DataComp, CC12M) and synthetic data generation (Section 2.4, 3.1). While standard for the field, the absence of a specific discussion on how Personally Identifiable Information (PII) was scrubbed, how copyright compliance was managed, or whether opt-out requests from data subjects were honored is a significant gap. Given the increasing regulatory scrutiny on AI training data (e.g., GDPR, EU AI Act), the authors should explicitly detail their data cleaning pipeline's approach to privacy and consent to ensure the model does not inadvertently memorize or reproduce sensitive user data.

Second, the **dual-use risks** associated with the "Agentic RL" capabilities (Section 3.4, Case VI) require careful mitigation. The model is demonstrated to autonomously book hotels, purchase goods, and execute payments. While presented as a feature, this functionality introduces risks of unauthorized financial transactions, fraud, or automated supply chain attacks if the model is compromised or misused. The manuscript should include a dedicated safety analysis discussing these risks and describing the specific guardrails, rate limits, or human-in-the-loop verification steps implemented to prevent malicious exploitation.

Finally, the inclusion of **medical and scientific reasoning** (Case III: Anatomical Reasoning) necessitates a clear disclaimer. The model's ability to identify errors in medical diagrams could be misinterpreted as diagnostic capability. To prevent potential harm from hallucinations or misinterpretations in critical domains, the authors must explicitly state that the model is not a medical device and should not be used for clinical decision-making.

Addressing these points will strengthen the paper's ethical standing and ensure responsible deployment of the technology.

Action items.

- [**writing**] The paper describes training on massive web-scraped datasets (LAION, DataComp) and synthetic data generation but lacks a dedicated section on data privacy, consent, and copyright compliance. Explicitly address how personally identifiable information (PII) was handled and whether opt-out mechanisms were respected to mitigate legal and ethical risks.
- [**writing**] The 'Agentic RL' section details capabilities for autonomous tool use, booking services, and payment execution (Case VI). The manuscript must include a safety analysis regarding potential misuse for unauthorized transactions, fraud, or supply chain attacks, and describe any guardrails or human-in-the-loop constraints implemented.
- [**writing**] The 'Anatomical Reasoning' case study (Case III) involves medical diagrams and the correction of scientific errors. The authors should clarify that the model is not a diagnostic tool and include a disclaimer to prevent users from relying on its outputs for medical decision-making, mitigating risks of harm from hallucinations in critical domains.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The manuscript presents a technically ambitious architecture (MoE + DSA) for long-context multimodal modeling, but the scientific evidence supporting the central claims of "lossless" scaling and "state-of-the-art" performance is currently insufficient due to a lack of statistical rigor and controlled ablation studies.

First, the performance claims in Table 1 (Section 4.1) rely on point estimates without measures of uncertainty. For instance, the reported 13.9% absolute gain on Video-MME-v2 (512 frames) over Qwen3.5 is substantial, but without standard deviations, confidence intervals, or significance testing (e.g., paired t-tests across multiple seeds), it is impossible to rule out random variance or benchmark-specific noise. In high-stakes model comparisons, reporting only mean scores is a methodological weakness that undermines the robustness of the "SOTA" claim.

Second, the core architectural claim of "lossless 256K context" (Abstract, Section 1) is not empirically substantiated. The paper demonstrates performance on benchmarks with up to 512 frames but does not provide a direct comparison against a dense-attention baseline at equivalent context lengths, nor does it show a degradation curve as context length increases. To validate the "lossless" assertion, the authors must demonstrate that the sparse attention mechanism (DSA) retains information critical for reasoning at 256K tokens that would be lost in standard subsampling or truncation strategies. The current evidence is correlational (high scores on long-video benchmarks) rather than causal (proving the mechanism preserves information).

Finally, the reported "1% improvement" from Video-RL (Section 3.2.4) is presented without a clear baseline or statistical context. A 1% gain on a specific subset of 31K samples could be a statistical fluke or a result of overfitting to the specific reward model (Qwen2.5-VL-72B) used for evaluation. The authors should clarify the baseline metric, the specific benchmark subset used for this claim, and whether the improvement holds across different evaluation seeds.

To strengthen the scientific validity of the report, the authors must include error bars or significance tests for all major benchmark comparisons, provide an ablation study isolating the effect of the DSA module on long-context retention, and fully detail the experimental setup for the RL improvements.

Action items.

- [**science**] Table 1 (Video-MME-v2) reports a 13.9 point gap between Keye-VL-2.0 (42.4) and Qwen3.5 (28.5) on 512-frame inputs, yet the text claims 'state-of-the-art' without reporting standard deviation or confidence intervals. Provide statistical significance testing (e.g., t-tests) or error bars to confirm this gain is not due to random variance.
- [**science**] The claim of 'lossless 256K context' (Abstract, Section 1) lacks empirical validation. The paper reports performance on 512-frame benchmarks but does not provide a controlled ablation study comparing 256K context performance against a dense-attention baseline or a subsampled baseline to prove the 'lossless' nature of the sparse aggregation.
- [**science**] The Video-RL section (Section 3.2.4) cites a '1% improvement' from 31K video samples but does not specify the baseline metric or the specific benchmark subset. Without the baseline score and the specific evaluation protocol, this effect size is unverifiable and risks

p-hacking interpretation.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The manuscript presents a comprehensive technical report on Kwai Keye-VL-2.0, detailing architectural innovations (DSA, MoE) and extensive benchmarking. However, from a statistical analysis perspective, the evaluation section relies heavily on point estimates without measures of uncertainty.

In **Table 1 (Video Understanding Evaluation)** and **Table 2 (Code Agent Evaluation)**, the authors report single scalar scores (e.g., 74.1 on LongVideoBench, 64.2 on LiveCodeBench). For large language models, inference results can vary significantly based on sampling temperature, random seeds, and the specific subset of the benchmark used. Without reporting confidence intervals (CIs), standard deviations, or results averaged over multiple independent runs, it is impossible to determine if the reported margins (e.g., the 3.6 point lead over InternVL3.5 on LongVideoBench) are statistically significant or merely artifacts of random variance. Standard practice in the field requires reporting $mean \pm std$ or 95% CIs for such claims.

Furthermore, in **Section 4.2.4 (Video RL)**, the authors state that training on 31K video samples "improves benchmarks by 1%." This claim is statistically ambiguous. A 1% absolute improvement on a benchmark with a score of ~80 is a small effect size. The manuscript does not provide the baseline variance, the specific benchmark subset used for this claim, or a statistical test (e.g., t-test or bootstrap) to validate that this improvement is distinguishable from noise.

Finally, **Table 1** includes a "Non-Lin" metric where Keye-VL-2.0 underperforms InternVL3.5 (18.5 vs 26.3). The text focuses almost exclusively on the "ACC" metric where the model leads, potentially engaging in "cherry-picking" of favorable statistics. A rigorous review would require an analysis of whether this degradation is statistically significant and an explanation of the trade-offs involved in the model's design choices that lead to this specific variance.

To strengthen the scientific validity of the claims, the authors should re-run evaluations with multiple seeds to generate variance estimates and report these alongside the means.

Action items.

- [**science**] Table 1 (Video Eval) and Table 2 (Code Eval) report single-point accuracy scores without confidence intervals or standard deviations. Given the stochastic nature of LLM inference and benchmark sampling, report 95% CIs or results over multiple seeds to establish statistical significance of the claimed margins (e.g., 74.1 vs 70.5 on LongVideoBench).
- [**science**] The claim of '1% improvement' from Video RL (Section 4.2.4) lacks statistical context. Specify the baseline score, the sample size (N=31K), and the variance of the metric to determine if this gain is statistically significant or within the noise floor of the evaluation

protocol.

- **[writing]** The 'Non-Lin' metric in Table 1 (Video-MME-v2) shows a performance drop for Keye-VL-2.0 compared to InternVL3.5. The text does not discuss the statistical significance of this degradation or potential confounding factors (e.g., specific video length distributions) that might explain the variance.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript demonstrates a generally high level of technical writing, with clear exposition of complex architectural components like DSA and MOPD. The flow between the abstract, introduction, and methodology is logical, and the use of bolding effectively highlights key contributions. However, several specific issues regarding terminology and structural consistency require attention to meet publication standards.

First, in the Introduction (Section 1), the authors use the phrase "temporal continuousness" to describe the preservation of video context. This is not standard academic terminology; "temporal continuity" or "temporal coherence" would be more precise and idiomatic. Similarly, the term "agentic intelligence" is used frequently (Abstract, Introduction) but remains undefined. While the context implies capabilities related to tool use and planning, a brief explicit definition would prevent ambiguity for readers unfamiliar with the specific connotation intended by the authors.

Second, in Section 1.3, the subsection titled "MQA-Style Lightning Indexer" introduces the term "Lightning" as if it were a proper noun or a specific named component. Unless "Lightning Indexer" is a previously established term in the cited literature (which is not immediately clear from the context), this capitalization is confusing. It is recommended to either cite the specific origin of this term or rephrase it to "Efficient Indexer" or "Fast Indexer" to maintain descriptive clarity.

Finally, the Case Study section (Section 5) exhibits structural inconsistencies in the provided LaTeX source. The cases appear to be out of order (e.g., Case VI is presented before Case II in the text chunks), and the \label commands (e.g., `agentic_cases`, `image_cases`) do not always align with the visual flow of the cases (Text, Image, Video, Agent). The authors should ensure the cases are presented in a logical numerical sequence (I–VI) and that all cross-references and labels are updated to reflect this correct ordering. Addressing these points will significantly enhance the readability and professional polish of the report.

Action items.

- **[writing]** In the Introduction, the phrase 'temporal continuousness' is non-standard and likely a misuse of 'continuity' or 'temporal coherence'. Replace with 'temporal continuity' to improve clarity and professional tone.

- **[writing]** The term ‘agentic intelligence’ appears in the Abstract and Introduction but is never explicitly defined. Given the paper’s focus on agent tasks, a brief definition or clarification of the specific capabilities implied by this term is necessary for reader understanding.
- **[writing]** In Section 1.3, the phrase ‘MQA-Style Lightning Indexer’ uses ‘Lightning’ as a proper noun without prior introduction or citation. If this is a specific component name, it should be introduced formally; otherwise, consider a more descriptive term like ‘Fast’ or ‘Efficient’.
- **[writing]** The Case Study section (Section 5) contains inconsistent formatting for case labels (e.g., ‘Case VI’ appears before ‘Case II’ in the provided text chunks). Ensure the case studies are ordered logically (I through VI) and that all \label and ef commands correspond to the correct sequence.